RAMDE - Engenharia Orientada a Requisitos e Modelos

Mestrado em Engenharia de Sistemas Computacionais Críticos

Lecture 07

Modeling Standards

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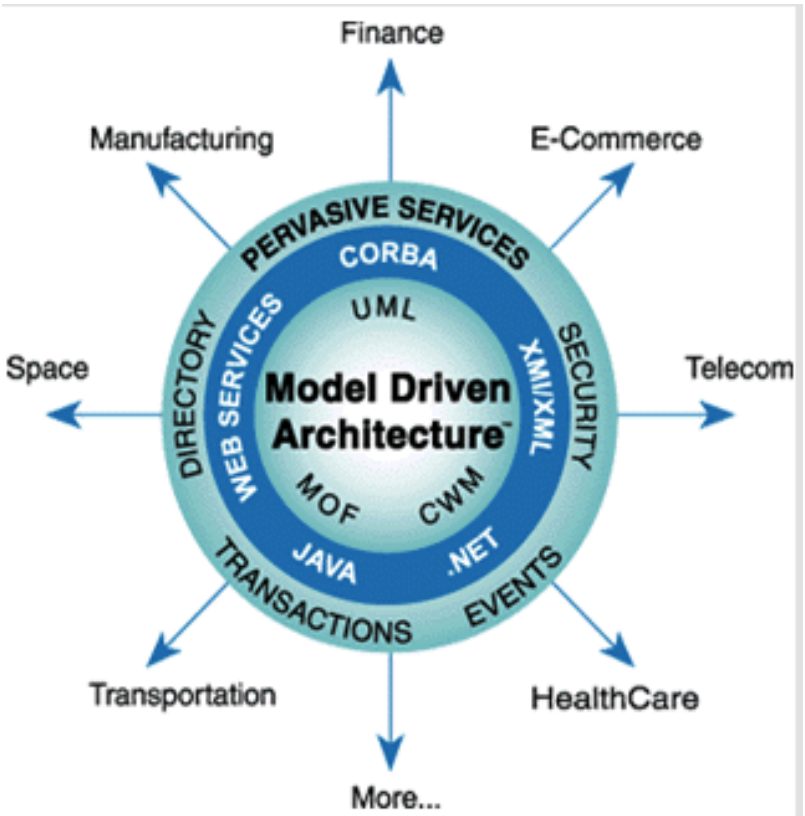
2026/2027

Contents

- Model-Driven Architecture (MDA) and other Standards
- Unified Modeling Language (UML)
- Systems Modeling Language (SysML)
- The Papyrus Modeling Tool
- Example

MDA Introduction

- **Model-driven Architecture (MDA[1])** is the particular vision of Model-Driven Engineering (MDE) proposed by the Object Management Group (OMG)

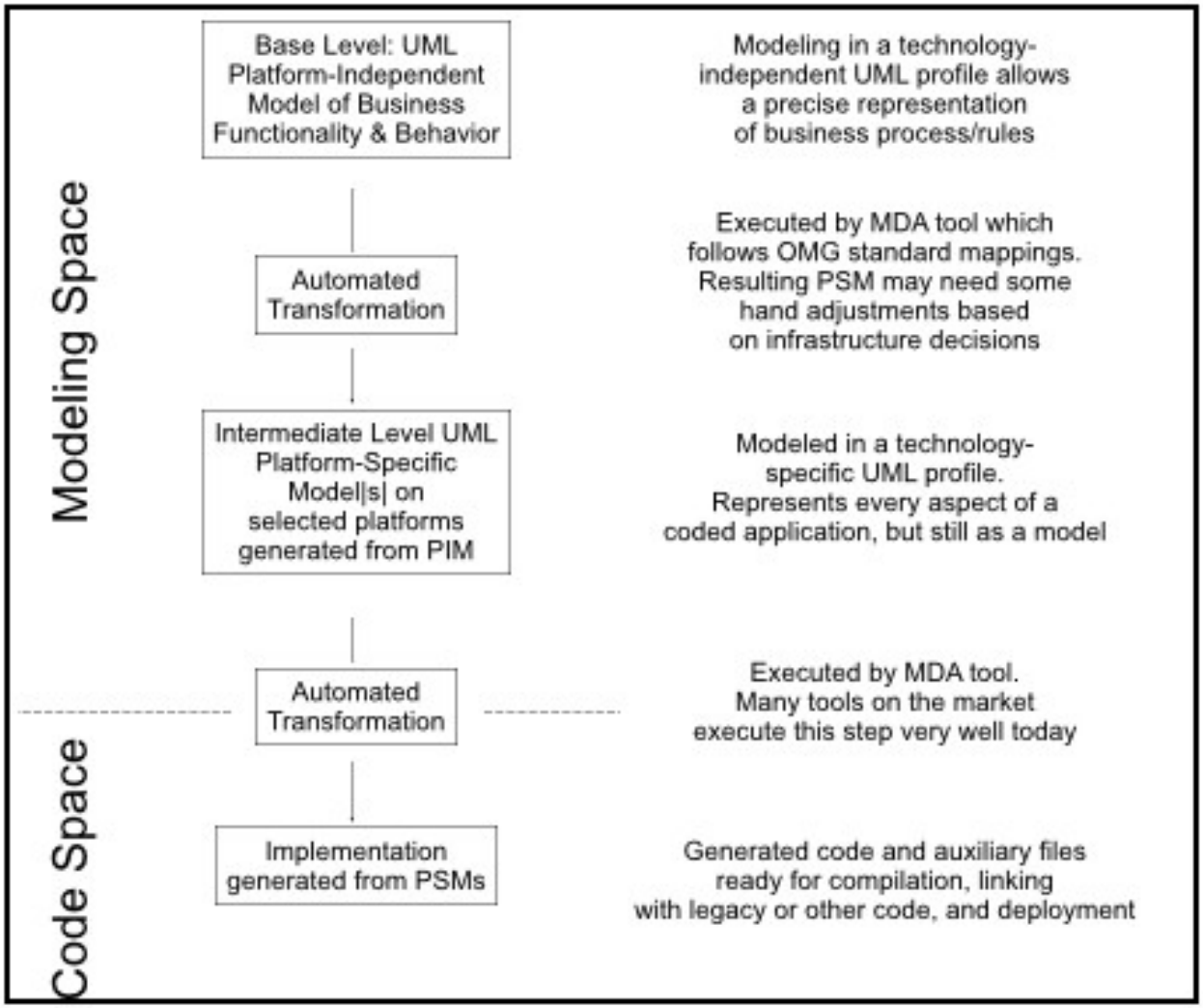


[1] http://www.omg.org/mda/

MDA Introduction

- **Interoperability** through platform independent models
  - Standardization initiative of the Object Management Group (**OMG**), based on OMG Standards, particularly **UML**
  - Counterpart to CORBA on the modeling level: interoperability between different platforms
  - Applications which can be installed on different platforms -> portability, no problems with changing technologies, integration of different platforms, etc.
- **Modifications to the basic architecture**
  - Segmentation of the model level
    - **Platform Independent** Models (PIM): valid for a set of (similar) platforms
    - **Platform Specific** Models (PSM): special adjustments for one specific platform
- Requires model-to-model transformation (PIM-PSM; compare QVT) and model-to-code transformation (PSM-Code)
- Platform development is not taken into consideration – in general industry standards like J2EE, .NET, CORBA are considered as platforms

MDA Development Cycle

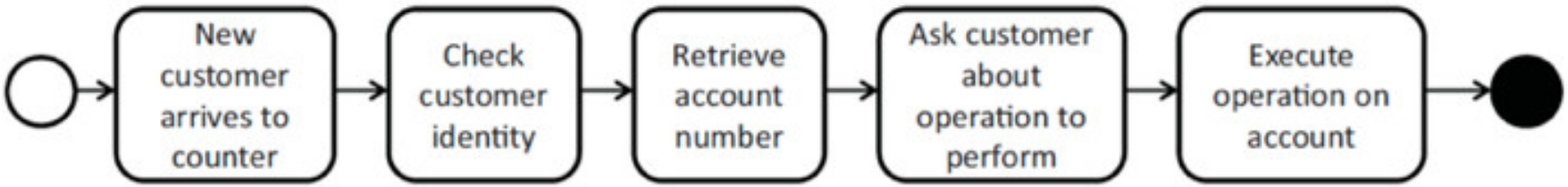


MDA Modeling Levels: CIM, PIM, PSM

- **Computation independent (CIM):** describe requirements and needs at a very abstract level, without any reference to implementation aspects (e.g., description of user requirements or business objectives);
- **Platform independent (PIM):** define the behavior of the systems in terms of stored data and performed algorithms, without any technical or technological details;
- **Platform-specific (PSM):** define all the technological aspects in detail.

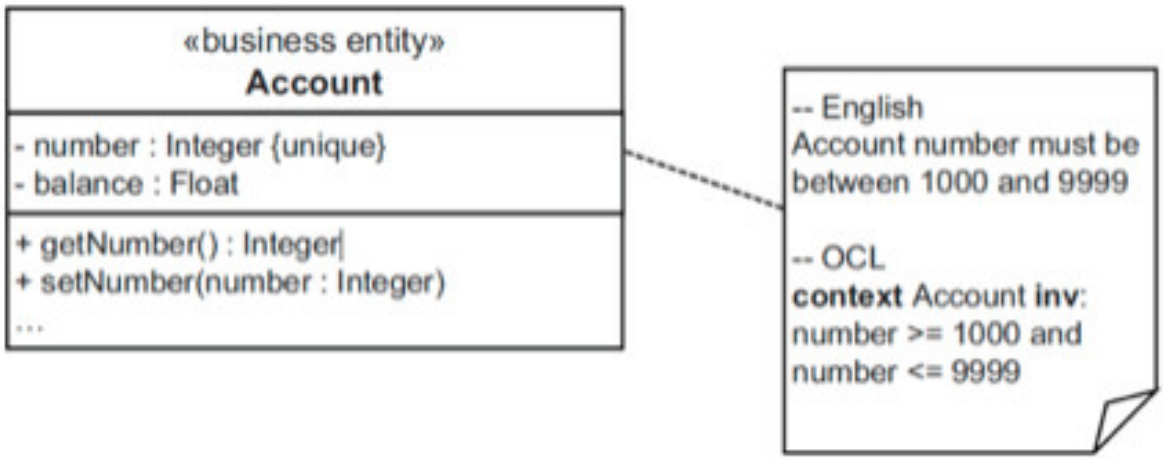
MDA Modeling Levels: CIM

- Eg., business process

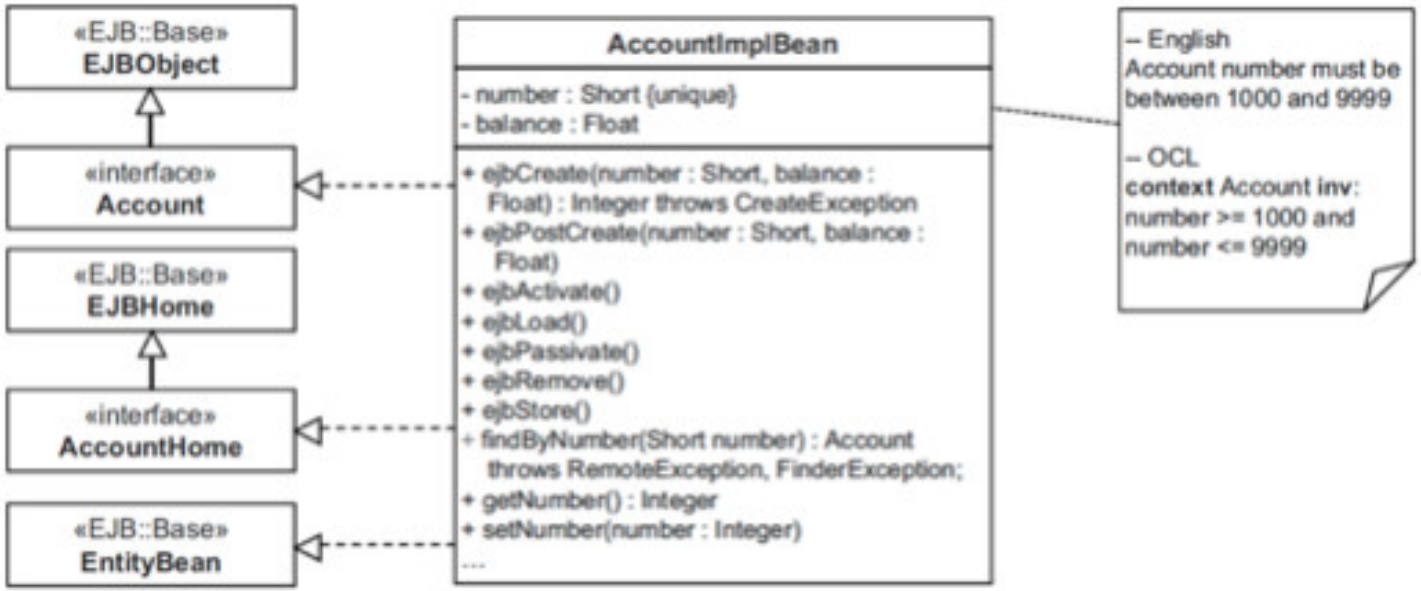


MDA Modeling Levels: MDA Platform Independent Model (PIM)

- Specification of structure and behaviour of a system, abstracted from technological details
- Using UML(optional)
- Abstraction of structure and behaviour of a system with the PIM simplifies the following:
  - Validation for correctness of the model
  - Create implementations on different platforms
  - Tool support during implementation



MDA Modeling Levels: MDA Platform Specific Model (PSM)

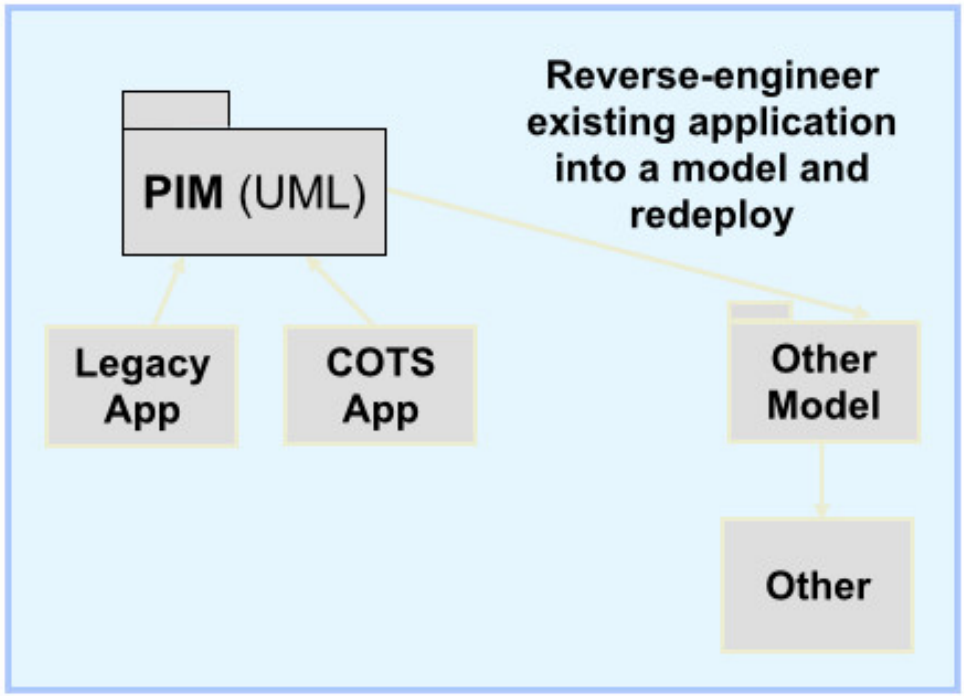


- Specifies how the functionality described in the PIM is realized on a certain platform
- Using a UML-Profile for the selected platform, e.g., EJB

MDA Approaches: MDA Reverse Engineering / Roundtrip Engineering

- Re-integration onto new platforms via Reverse Engineering of an existing application into a PIM and subsequent code generation

- MDA tools for Reverse Engineering automate the model construction from existing code



MDA: Other MDA Related OMG Standards

- CORBA - Common Object Request Broker Architecture
  - Language -and platform- neutral interoperability standard (similar to WSDL, SOAP and UDDI)
  - `https://www.omg.org/spec/CORBA/`
- UML - Unified Modeling Language
  - Standardized modeling language, industry standard
  - `https://www.omg.org/spec/UML/`
- CWM - Common Warehouse Metamodel
  - Integrated modeling language for Data Warehouses
- MOF – Meta Object Facility
  - A standard for metamodels and model repositories
  - `https://www.omg.org/spec/MOF/`
- XMI - XML Metadata Interchange
  - XML-based exchange of models
  - `https://www.omg.org/spec/XMI/`
- QVT – Queries/Views/Transformations
  - Standard language for Model-to-Model transformations
  -
- UML Profiles
  - UML Profiles tailor the language to particular areas of computing (such as Enterprise Distributed Object Computing) or particular platforms (such as EJB or CORBA)
- ...

MDA: Other MDA Related OMG Standards

- **Systems engineering** is a multi-disciplinary approach to system development that is intended to ensure the pieces of the system work together to achieve the objectives of the whole. This requires architecting balanced solutions that satisfy diverse stakeholder needs for capability, dependability, sustainability, ease of use and affordability while managing complexity and risk.
- **Systems engineers** have traditionally captured information about the system in a variety of documentation including text, informal diagrams and spreadsheets. **The transition to model-based systems engineering (MBSE) captures this information in a system model which can improve the completeness, consistency, precision and traceability of the information**. This in turn can significantly reduce errors that often surface later in the development lifecycle causing cost and schedule overruns and unmet stakeholder expectations.

- **Systems Modeling Language (SysML)**
  - **A general-purpose modeling language for systems engineering** whose intent is to specify the language so that systems engineering modelers may learn to apply and use SysML, modeling tool vendors may implement and support SysML, and both can provide feedback to improve future versions.
  - `https://www.omg.org/spec/SysML/`

Some (more specific) Industry Standards

- CEN ISO/IEEE 11073, a standard focused on medical device communication - https://youtu.be/7jYzd9o0680
- ISO/SAE 21434 Road vehicles - Cybersecurity engineering and ISO/DPAS 5112 Road vehicles - Guidelines for auditing cybersecurity engineering - https://youtu.be/J1mzZpVgNkM
- ISO 26262 Road Vehicles - Functional safety and ISO/PAS 21448 Road vehicles - Safety of the intended functionality - https://youtu.be/5gZZJgIIvGM
- More videos availabe at https://www.youtube.com/channel/UCBvhaW8hkWgopiJWbFBrIFQ/featured

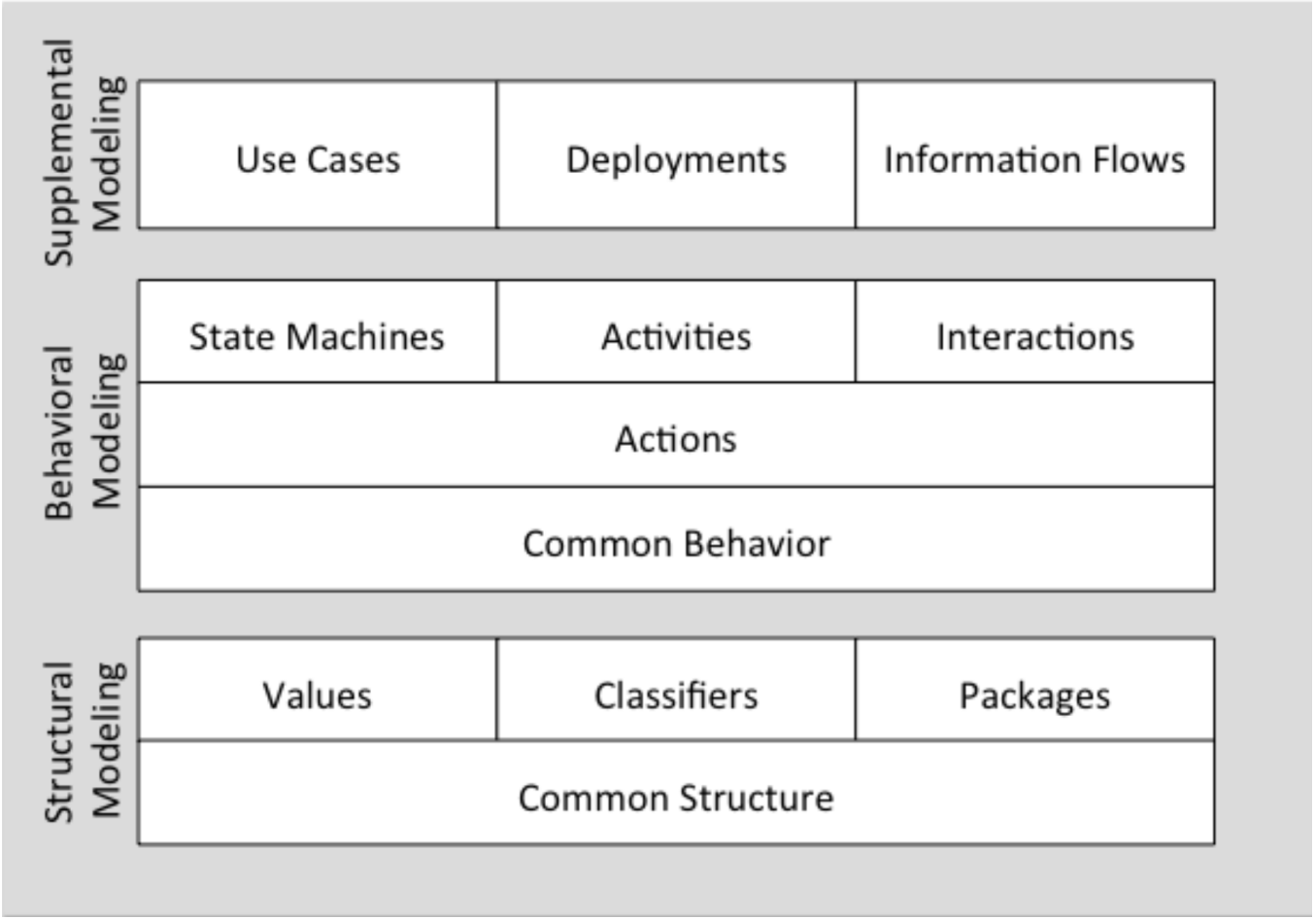
MDA: Discussion

- Problems when using **UML** as PIM/PSM
  - Method bodies?
  - Incomplete diagrams, e.g. missing attributes
  - Inconsistent diagrams
  - *For the usage of the UML in Model Engineering special guidelines have to be defined and adhered to*
- Different requirements to **code generation**
  - get/set methods
  - Serialization or persistence of an object
  - Security features, e.g. Java Security Policy
  - *Using adaptable code generators or PIM-to-PSM transformations*
- **Expressiveness** of the UML
  - UML is mainly suitable for "generic" software platforms like Java, EJB, .NET
  - Lack of support for user interfaces, code, etc.
  - *MDA tools often use proprietary extensions*

MDA: Discussion

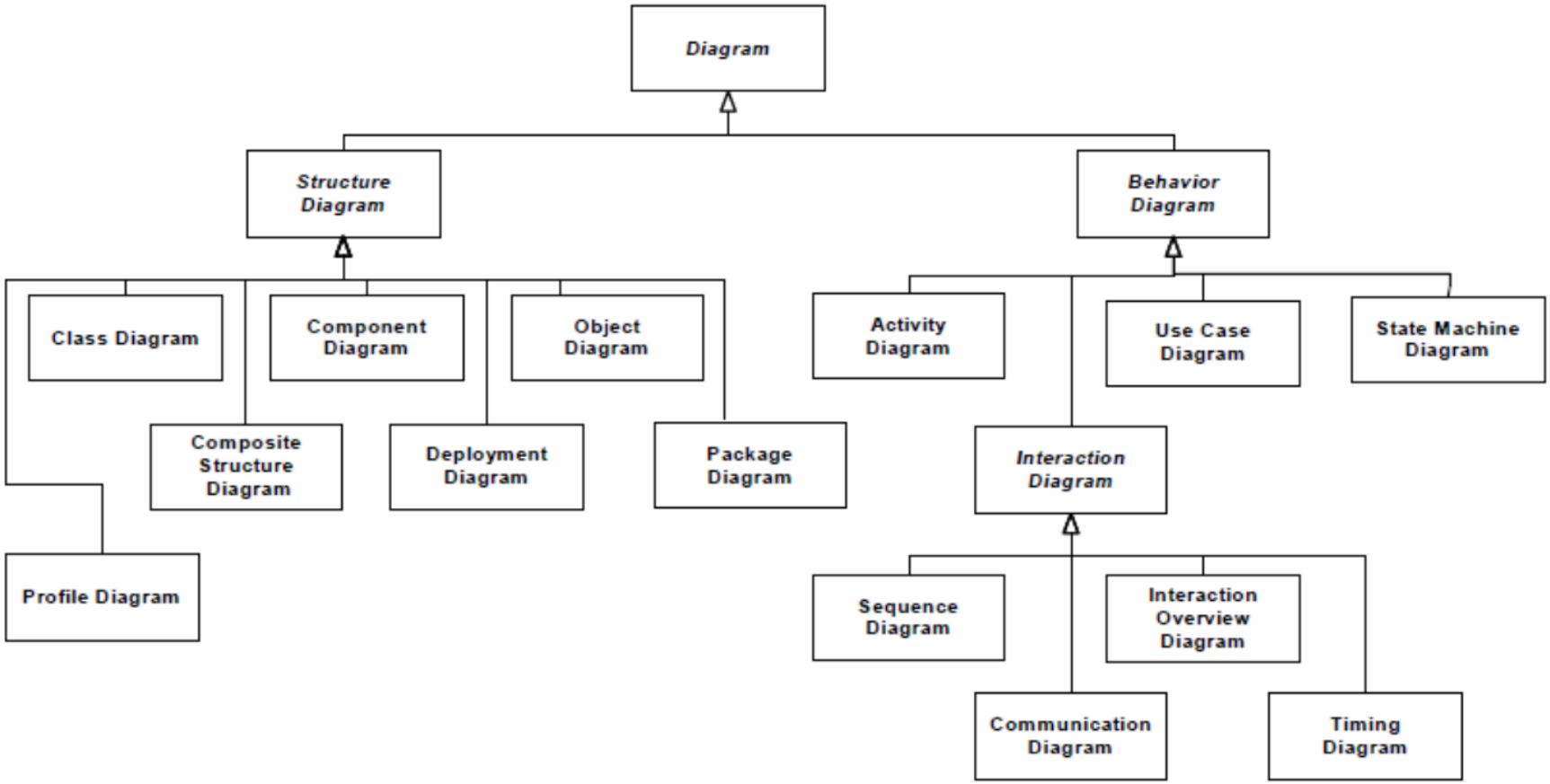
- Many **UML tools** are expanded to MDA tools
  - UML profiles and code generators
  - Stage of development partly still similar to CASE: proprietary UML profiles and transformations, limited adaptability
- **Advantages** of MDA
  - Standardization of the Meta-Level
  - Separation of platform independent and platform specific models (reuse)
- **Disadvantages** of MDA
  - No special support for the development of the execution platform and the modeling language
  - Modeling language practically limited to UML with profiles
  - Therefore limited code generation (typically no method bodies, user interface)

- The objective of UML is to provide system architects, software engineers, and software developers with tools for analysis, design, and implementation of **software-based systems as well as for modeling business and similar processes**.



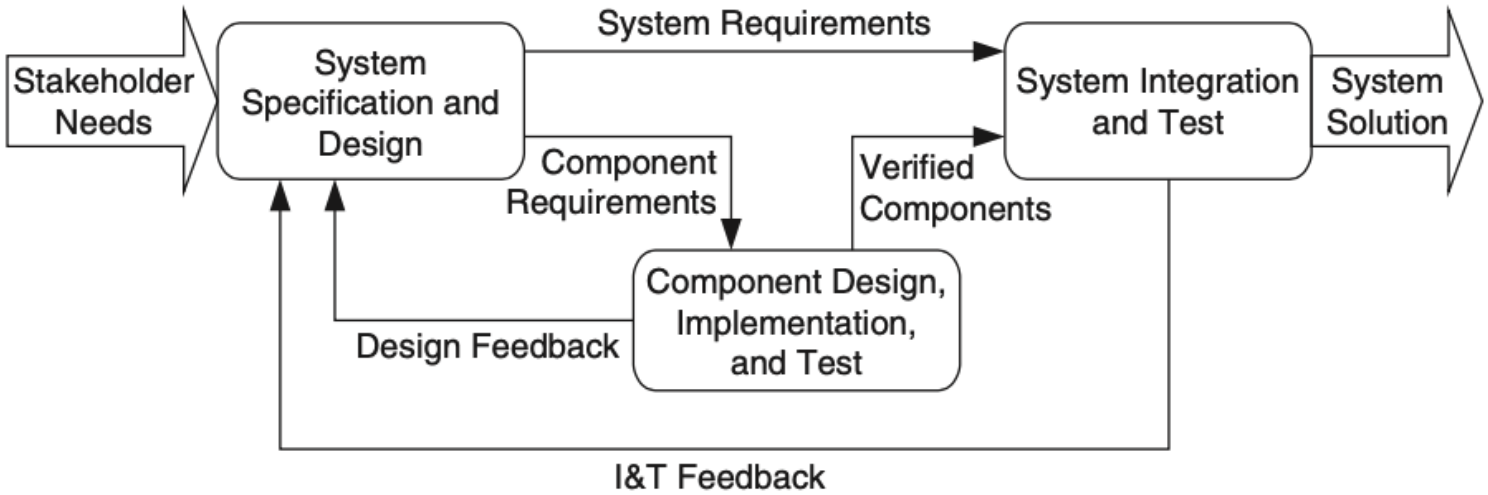
UML Diagrams

- UML Diagrams focus on software development

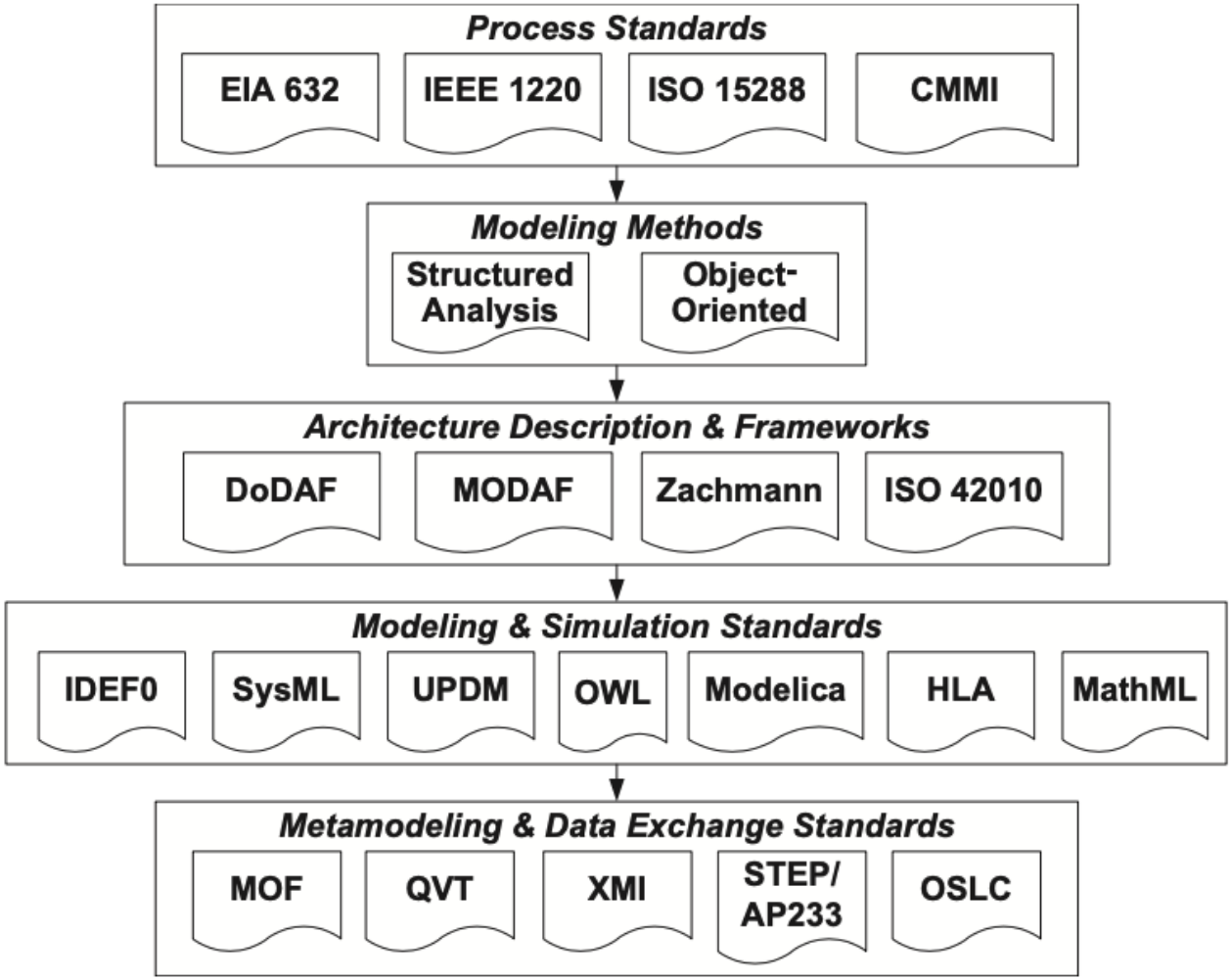


SysML

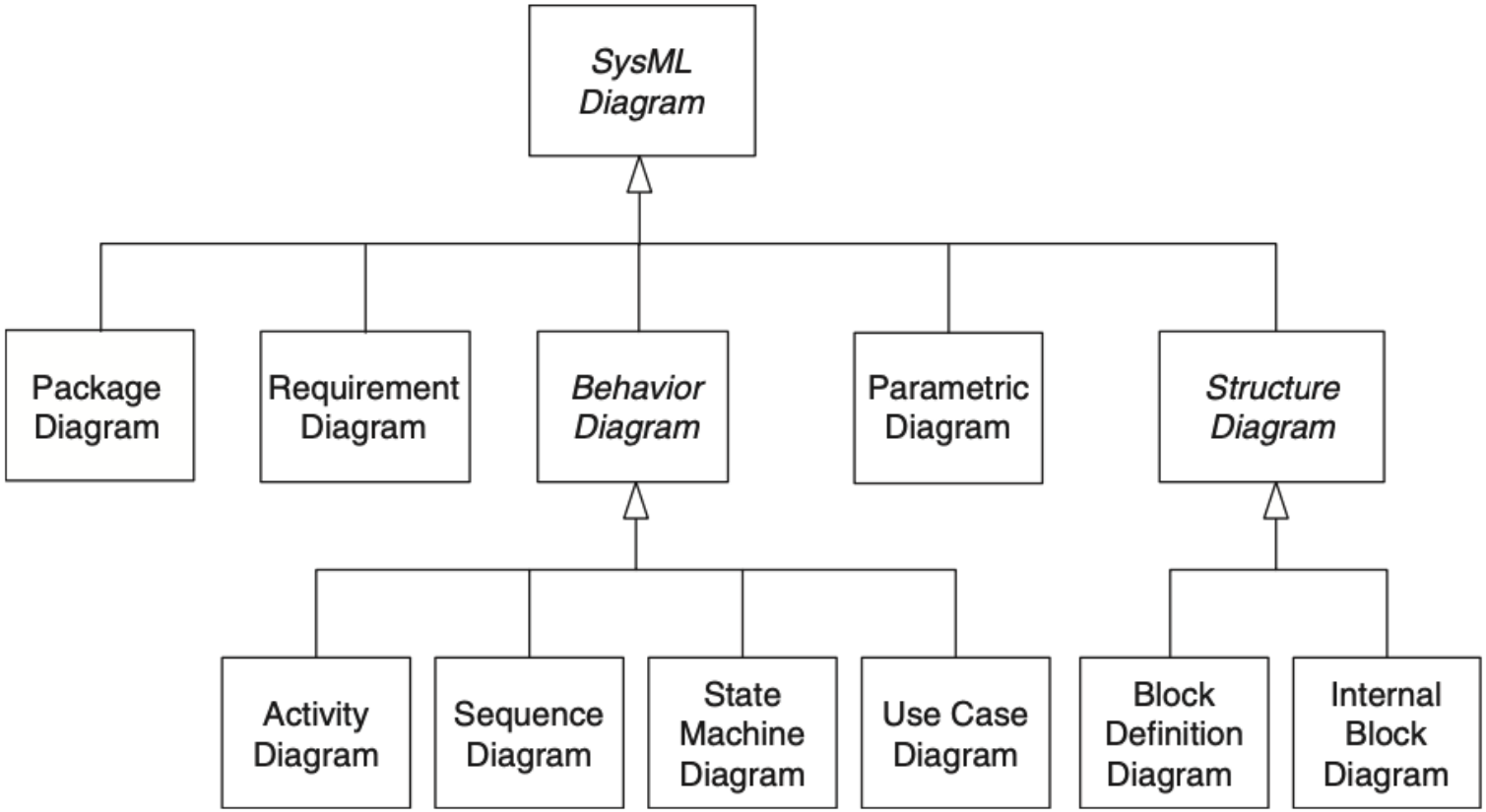
- The Object Management Group's OMG SysML is a general-purpose graphical modeling language for representing systems that may include **combinations of hardware and equipment, software, data, people, facilities, and natural objects**.
- SysML supports the practice of model-based systems engineering (MBSE) that is used to develop system solutions in response to complex and often technologically challenging problems.
- Systems Engineering Technical Processes:



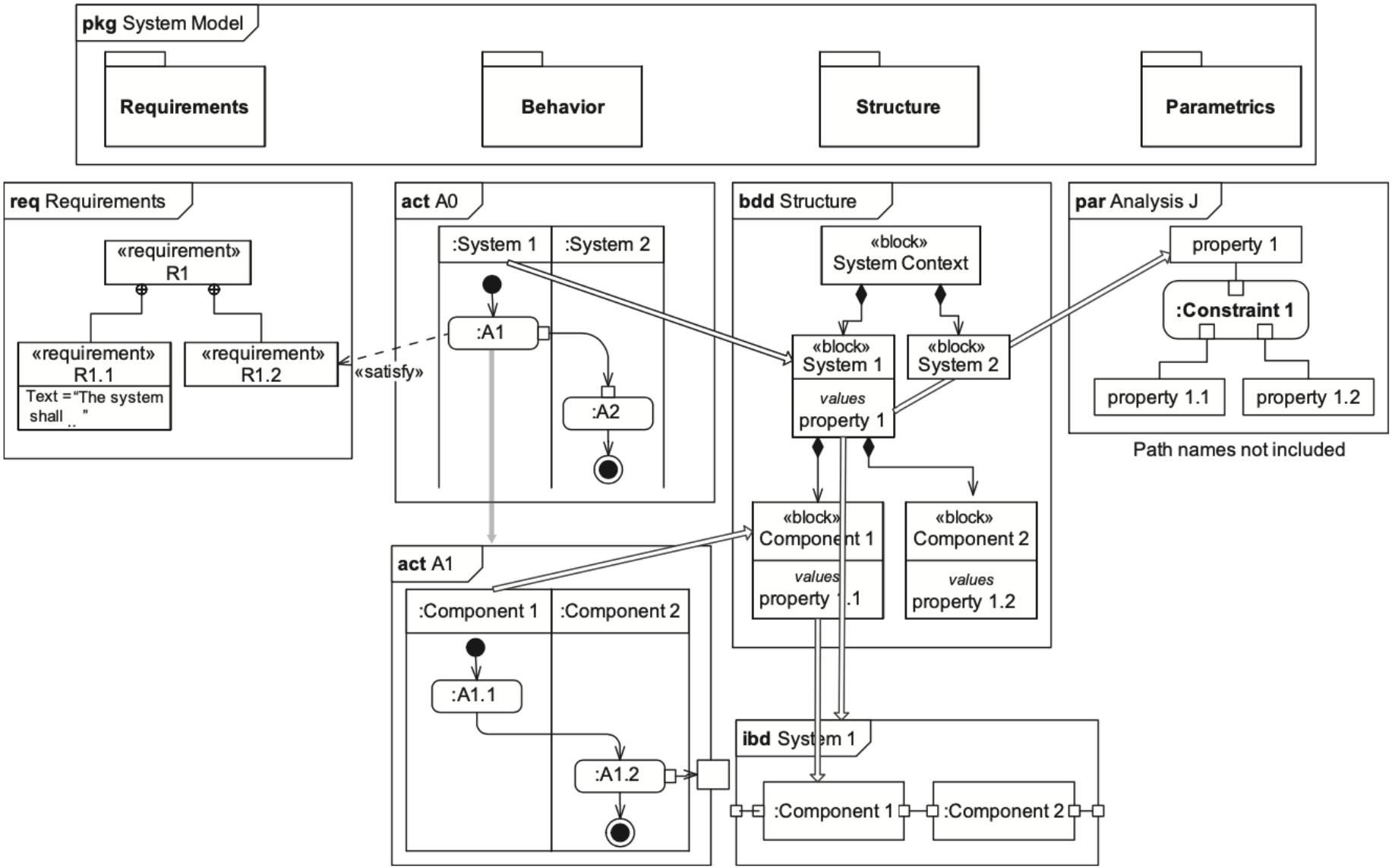
SysML and Other Standards



SysML Diagrams



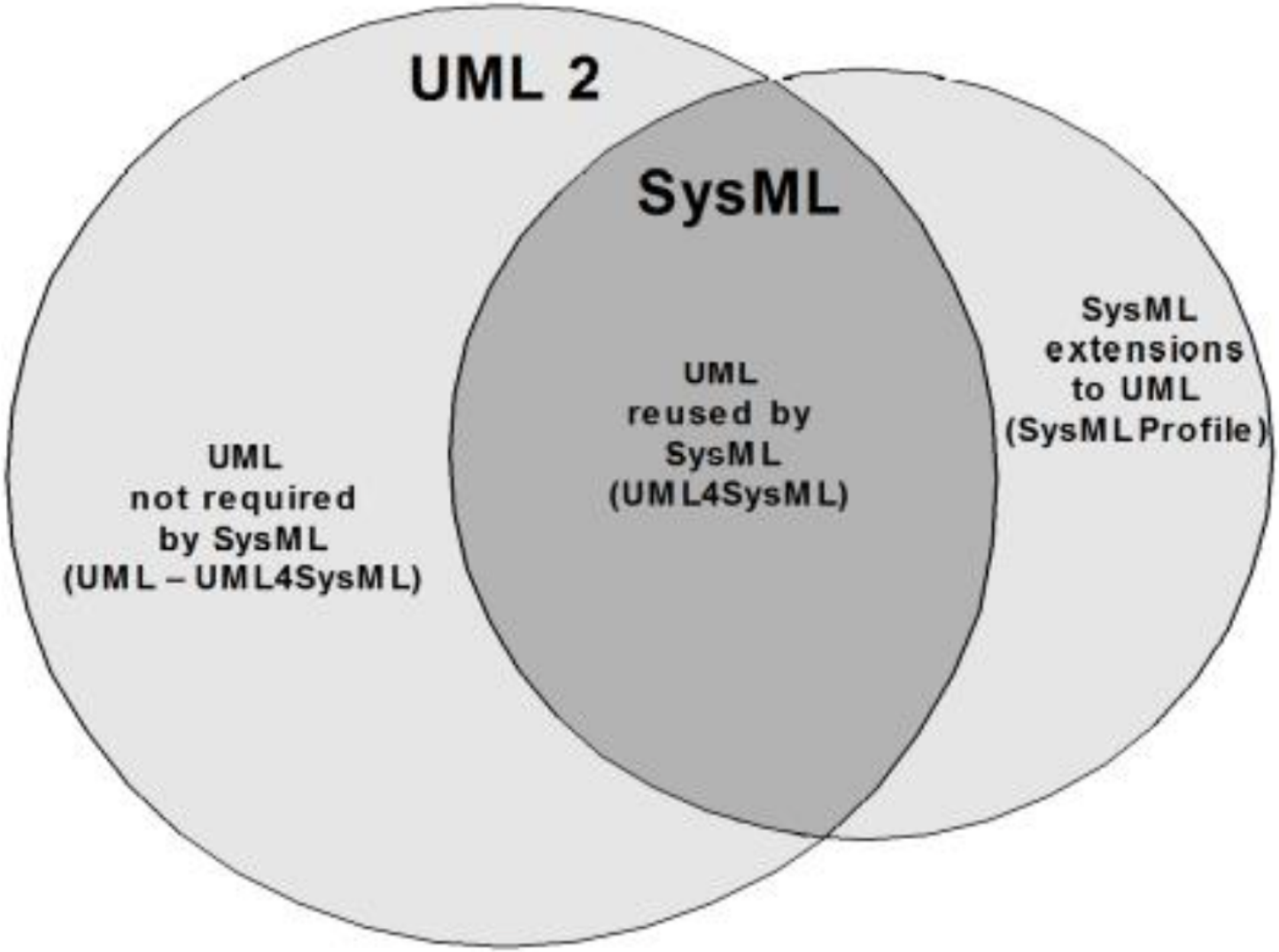
SysML Example



Tools: Papyrus

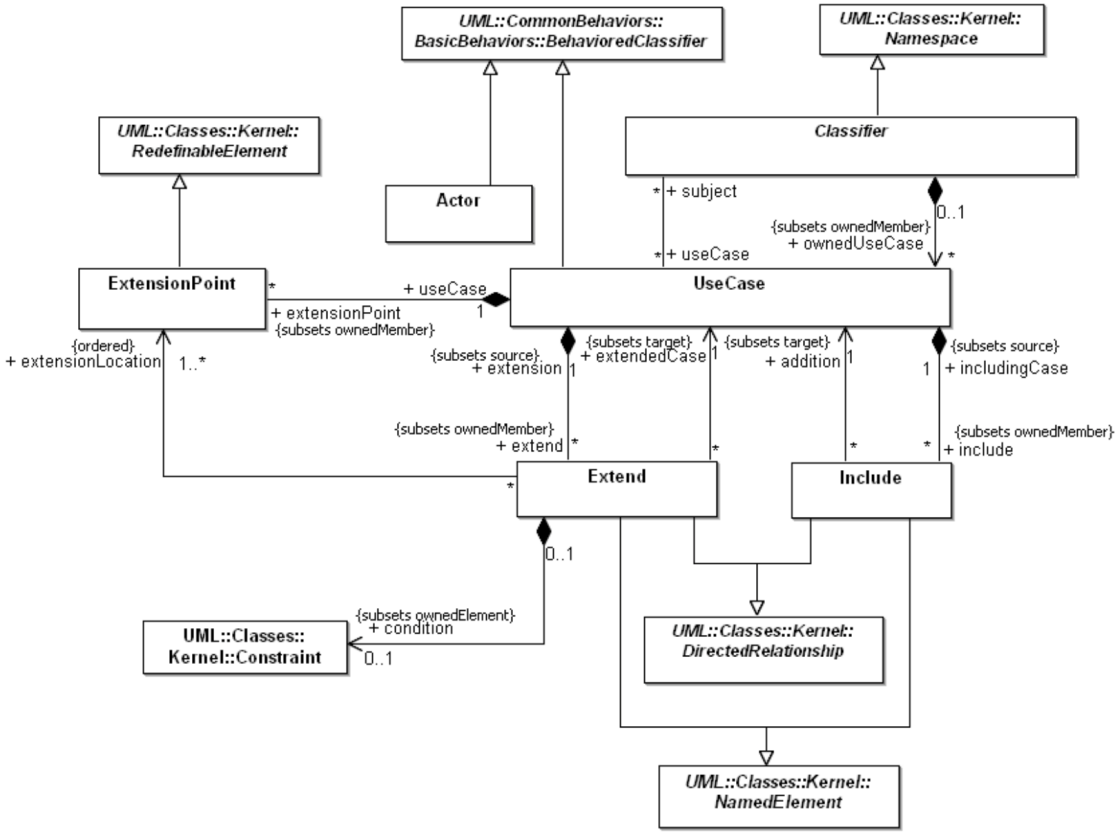
- Eclipse Papyrus is an industrial-grade open source Model-Based Engineering tool.
- Eclipse Papyrus has notably been used successfully in industrial projects and is the base platform for several industrial modeling tools.
- It implements standards such as UML and SysML
- `https://www.eclipse.org/papyrus/`

UML 2 and SysML 1.6 relationship



UML Metamodel

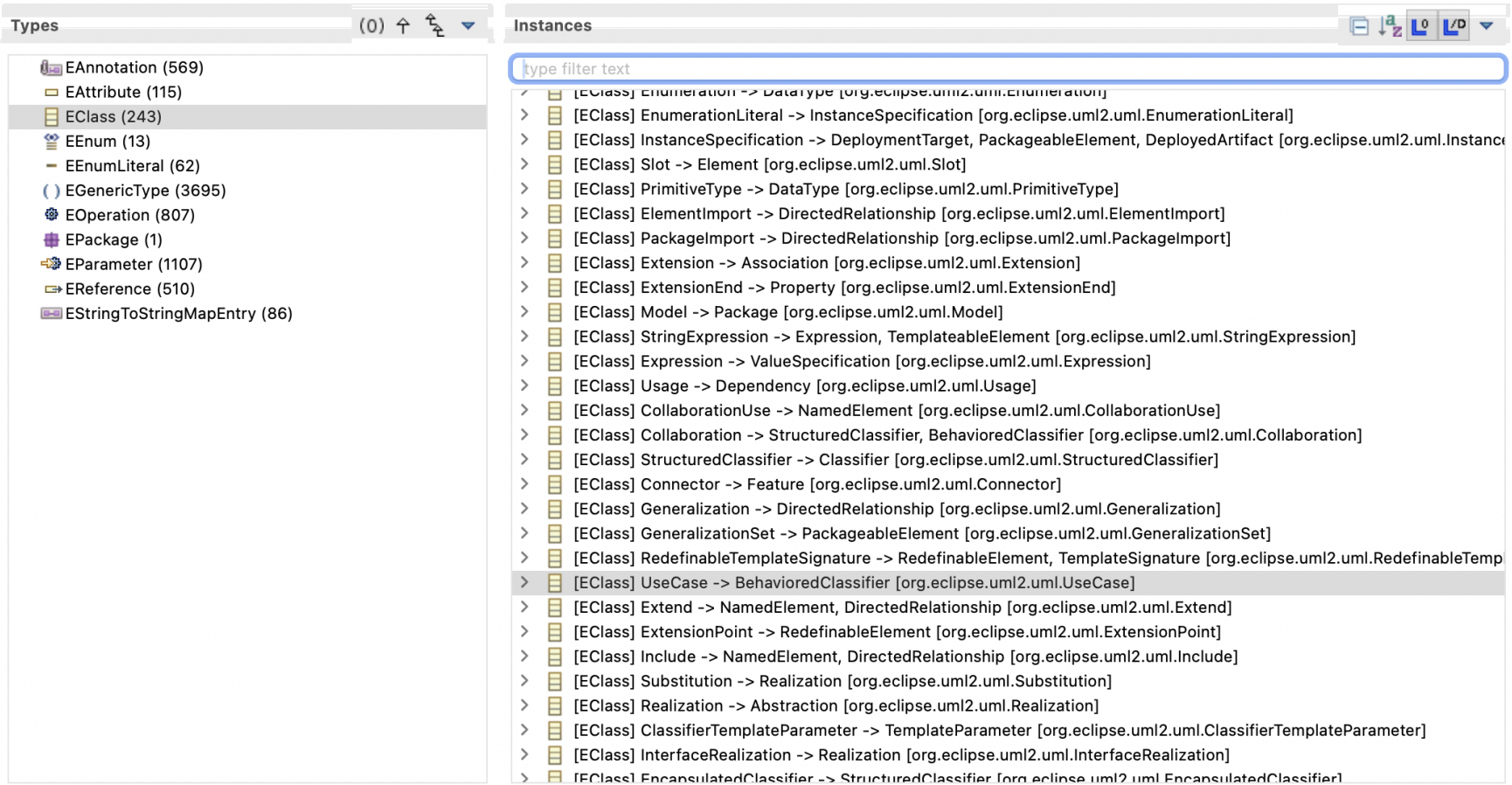
Partial excerpt of UML metamodel relating to use cases...



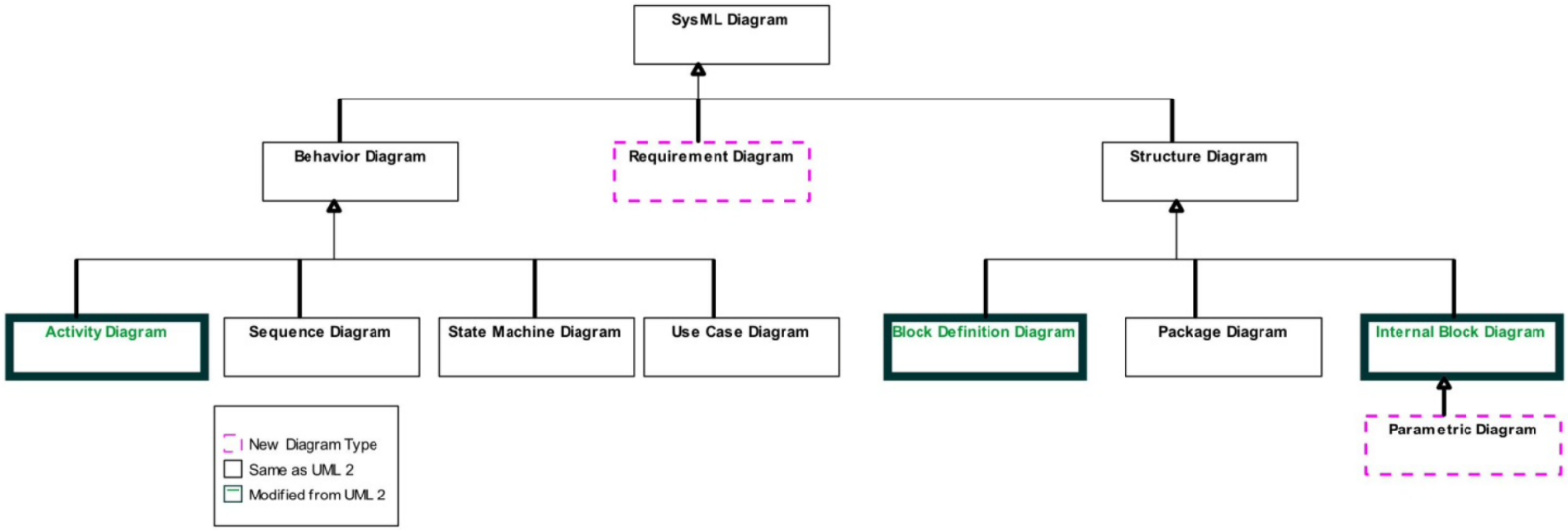
- The UML metamodel contains 243 classes....

UML Metamodel

UML metamodel implementation in Eclipse
(UML2, URI http://www.eclipse.org/uml2/5.0.0/UML)



SysML Metamodel



- SysML 1.6 extends UML via UML Profiles
- Number of SysML Stereotypes: 102 stereotypes (across SysML 1.4 to 1.6).

SysML Metamodel

SysML key stereotype extensions:

SysML Concept	**Underlying UML2 Metaclass**
Block	uml::Class
Requirement	uml::Class
Constraint Block	uml::Component or uml::Class
Value Type	uml::DataType
Proxy Port / Full Port	uml::Port
Flow Property	uml::Property
Satisfy / Verify / Refine	uml::Abstraction / uml::Dependency
Item Flow	uml::InformationFlow

SysML V2

SysML is evolving into SysML V2.

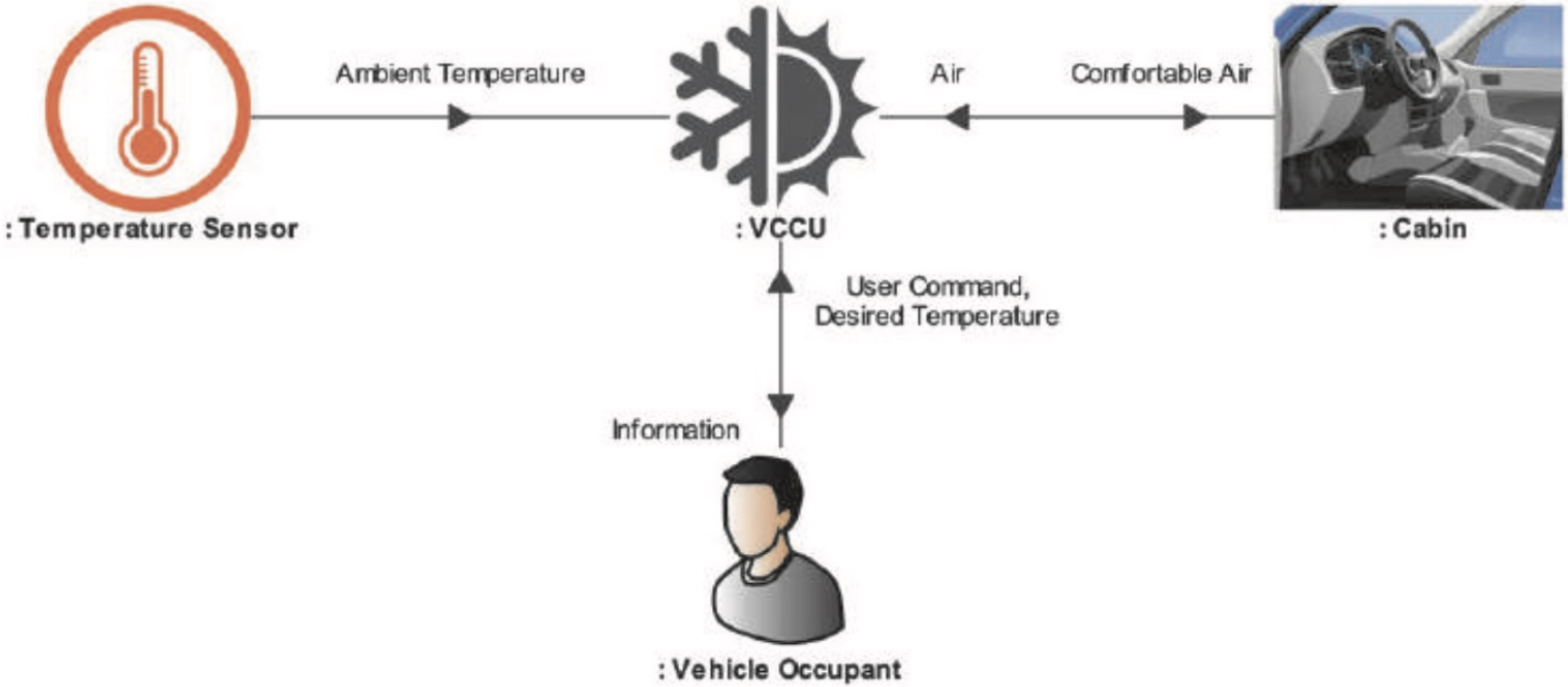
SysML v2 is the next-generation systems modeling language for Model-Based Systems Engineering (MBSE), offering a **human-readable textual notation** alongside **graphical syntax**, **formal semantics based on the Kernel Modeling Language** (KerML), and a **standardized REST API**.

- KerML Metamodel: Contains approximately 160 metaclasses.
- SysML v2 Metamodel: Extends KerML by adding approximately 150 additional specialized metaclasses.

KerML is implemented in MOF (The meta-metamodel used by Eclipse/EMF).

SysML Example - Vehicle Climate Control System (VCCS)

- We will follow the Vehicle Climate Control System example presented in the book "**MagicGrid Book of Knowledge, A Practical Guide to Systems Modeling using MagicGrid from Dassault Systemes**"
- For instructions on how to operate Papyrus see "Papyrus Tutorial, A Practical Guide to Systems Modeling with Papyrus and SysML"



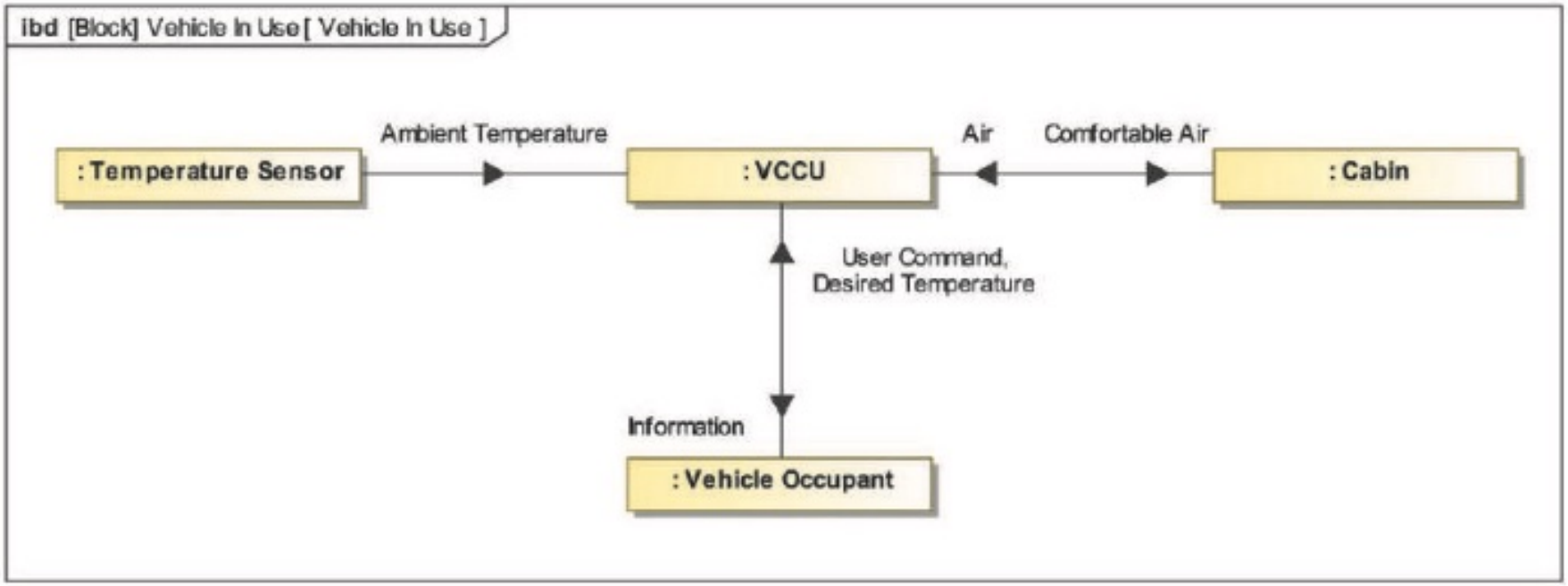
Example

- Black-Box Perspective: Capture the Stakeholder's needs in a Requirements Table. SysML supports diverse representations for its models. For instance, one can use tables or diagrams to represent requirements.

#	Name	Text
1	Ambient Temperature	I want to see the ambient temperature on the screen or some other output device.
2	Comfortable Temperature	I'd like to feel comfortable temperature while being in the cabin.
3	Desired Temperature	It should be a possibility to easily specify the desired temperature.
4	Heating & Cooling	The unit must be able to heat and cool.
5	Energy Consumption	I'd like the unit to consume as little energy as possible.
6	Manual Control	I should be able to start and stop climate control by myself.
7	Sound Level	Climate control unit in max mode shall not be louder than engine.
8	Total Mass	Mass of the unit shall not exceed 2 percent of the total car mass.

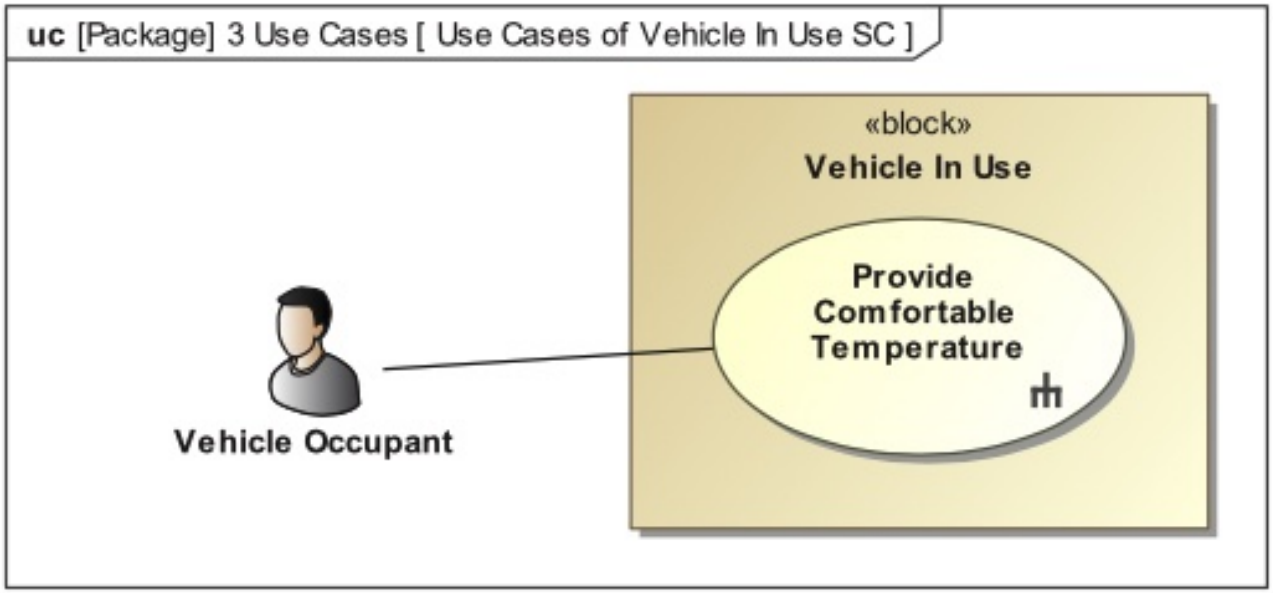
Example

- Black-Box Perspective: Model the system context
- At the center of the diagram you should represent the **SoI**, System of Interest (in this example the VCCU, Vehicle Climate Control Unit)



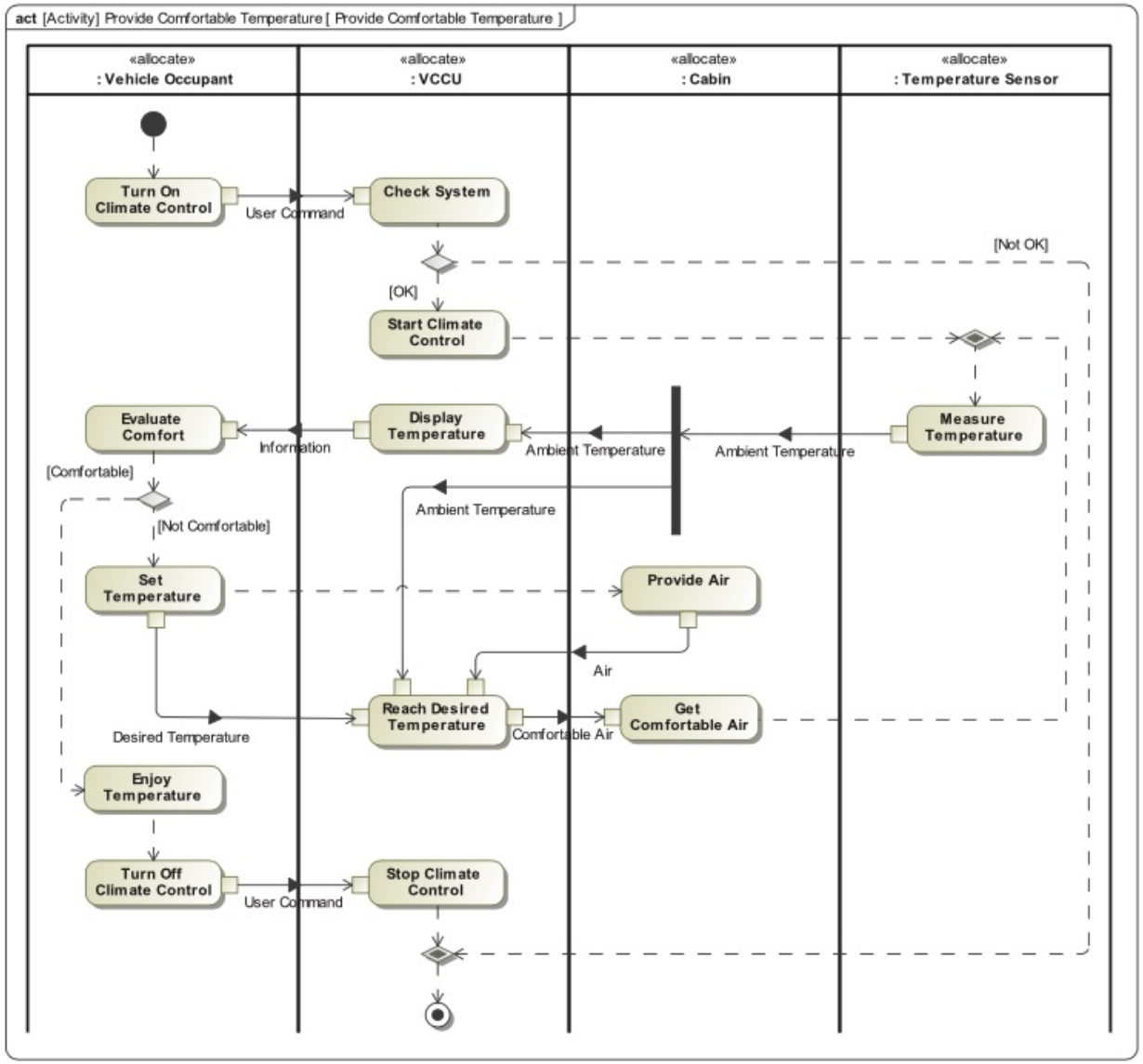
Example

- Black-Box Perspective: Model the use case



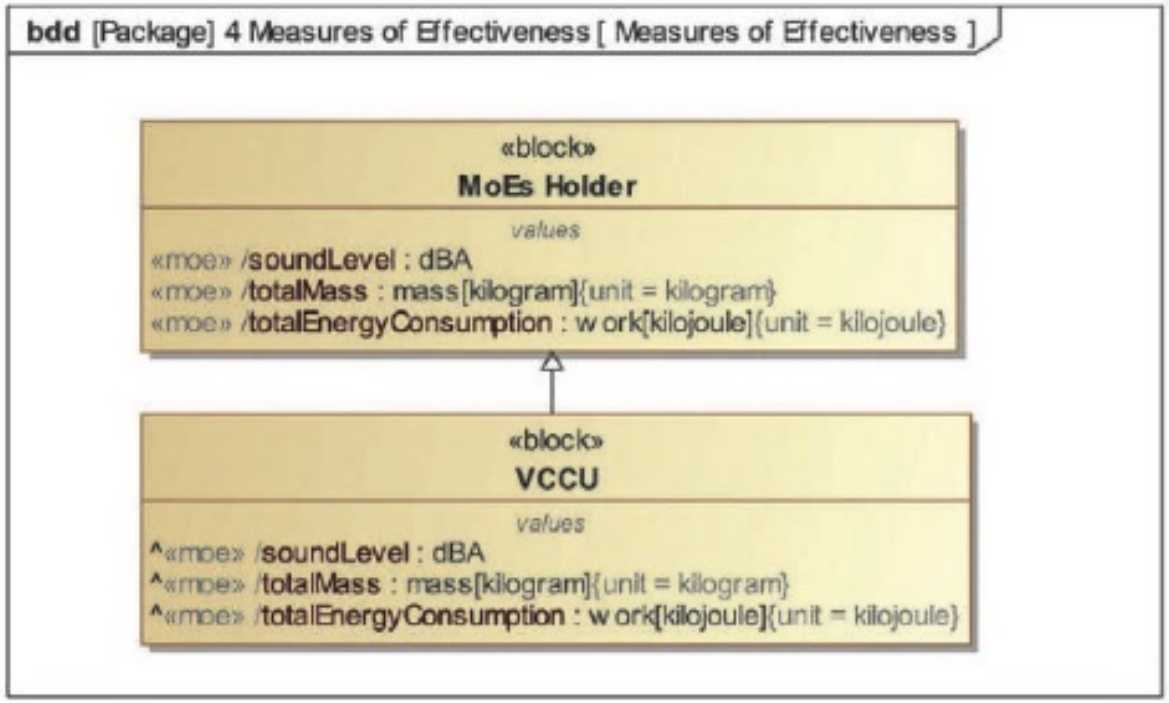
Example

- Black-Box Perspective: Model the main scenario of the use case (activity diagram)



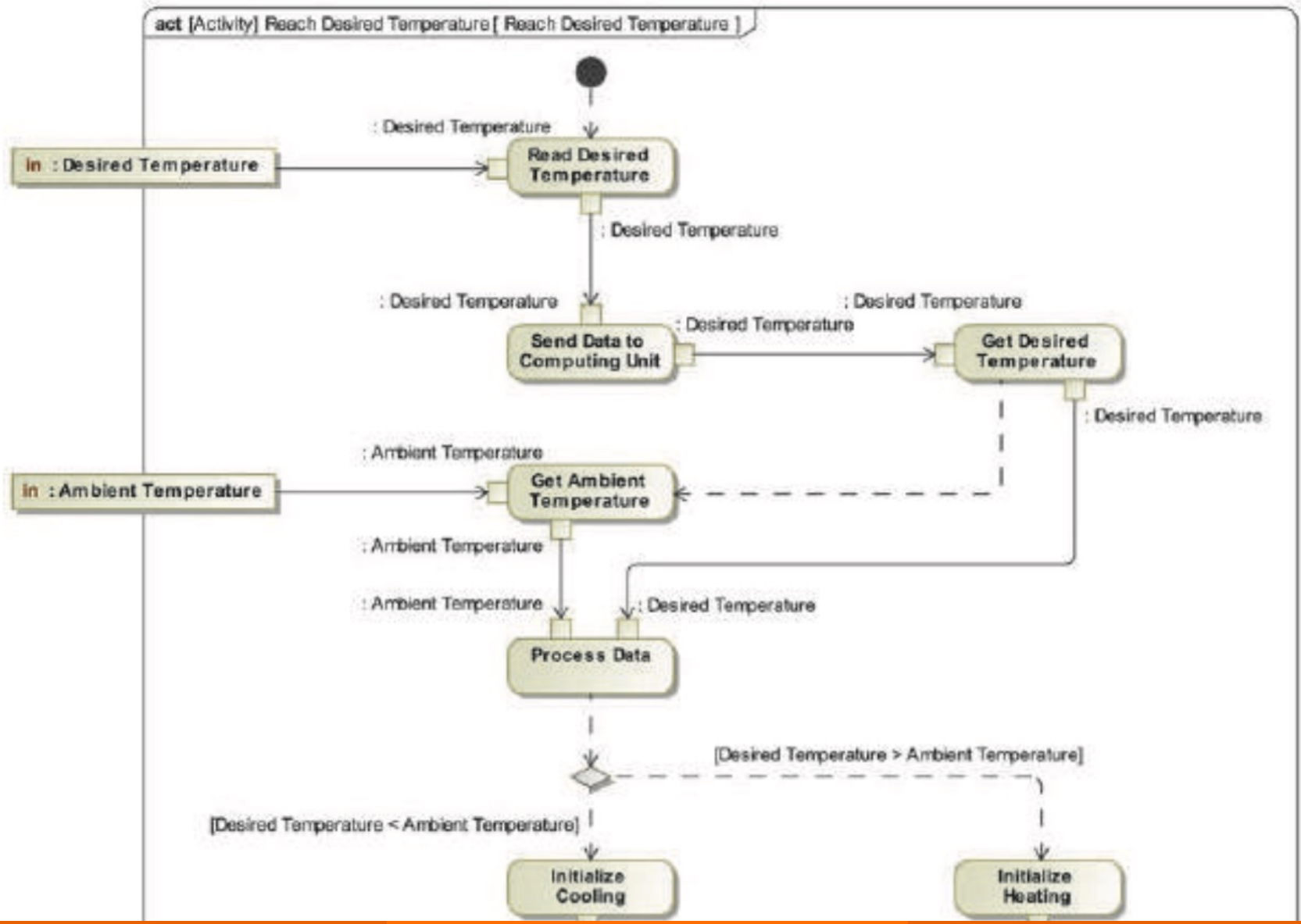
Example

- Black-Box Perspective: Model Measures of Effectiveness (MoEs)
- The non-functional stakeholder needs are refined with measures of effectiveness (MoEs), which capture quantifiable characteristics of the SoI in a numerical format



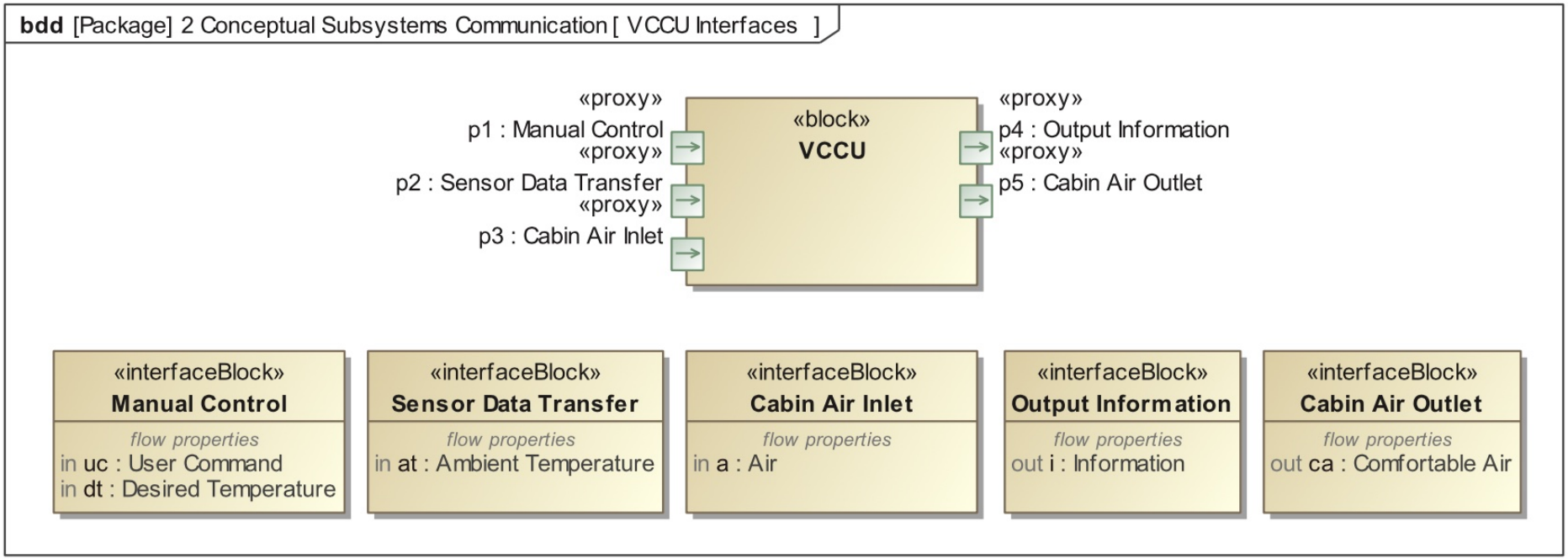
Example

- White-Box Perspective: Decompose a function



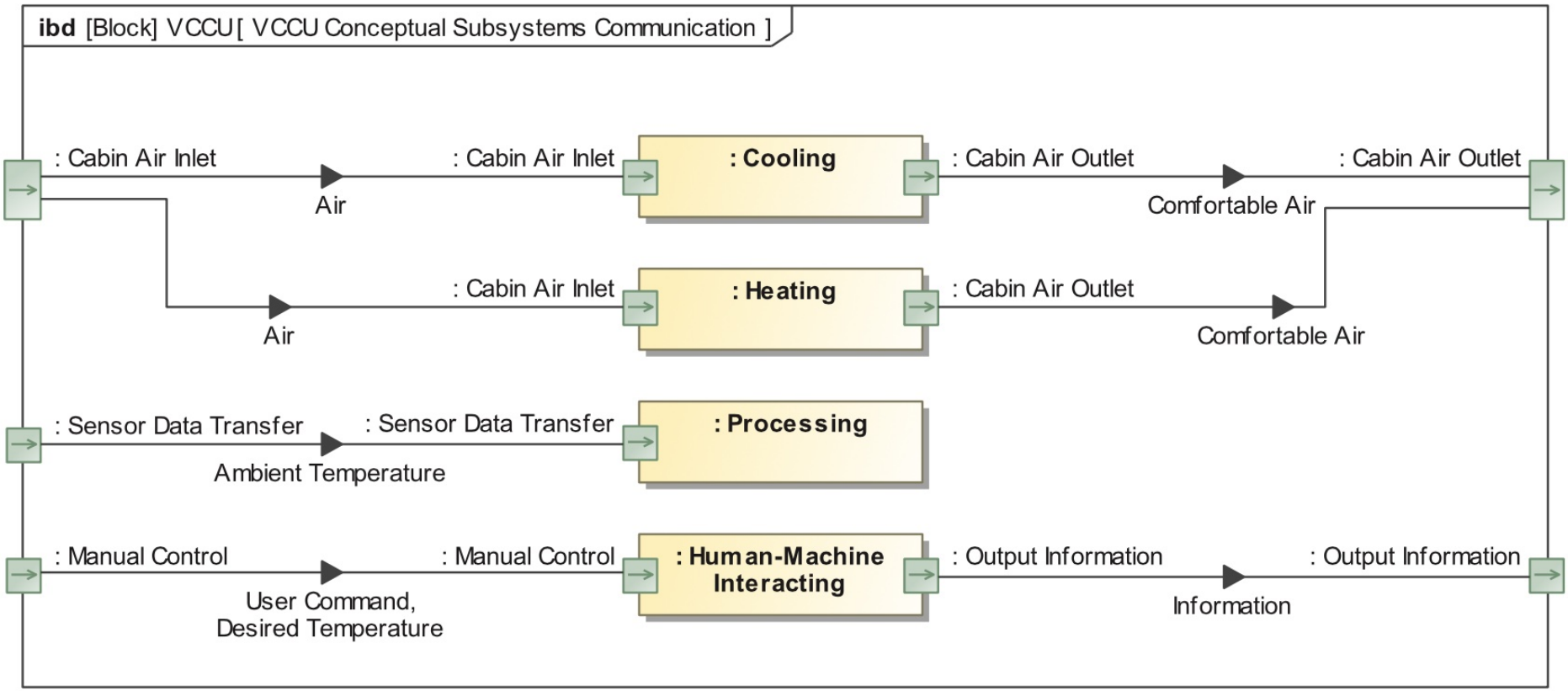
Example

- White-Box Perspective: Conceptual Interfaces



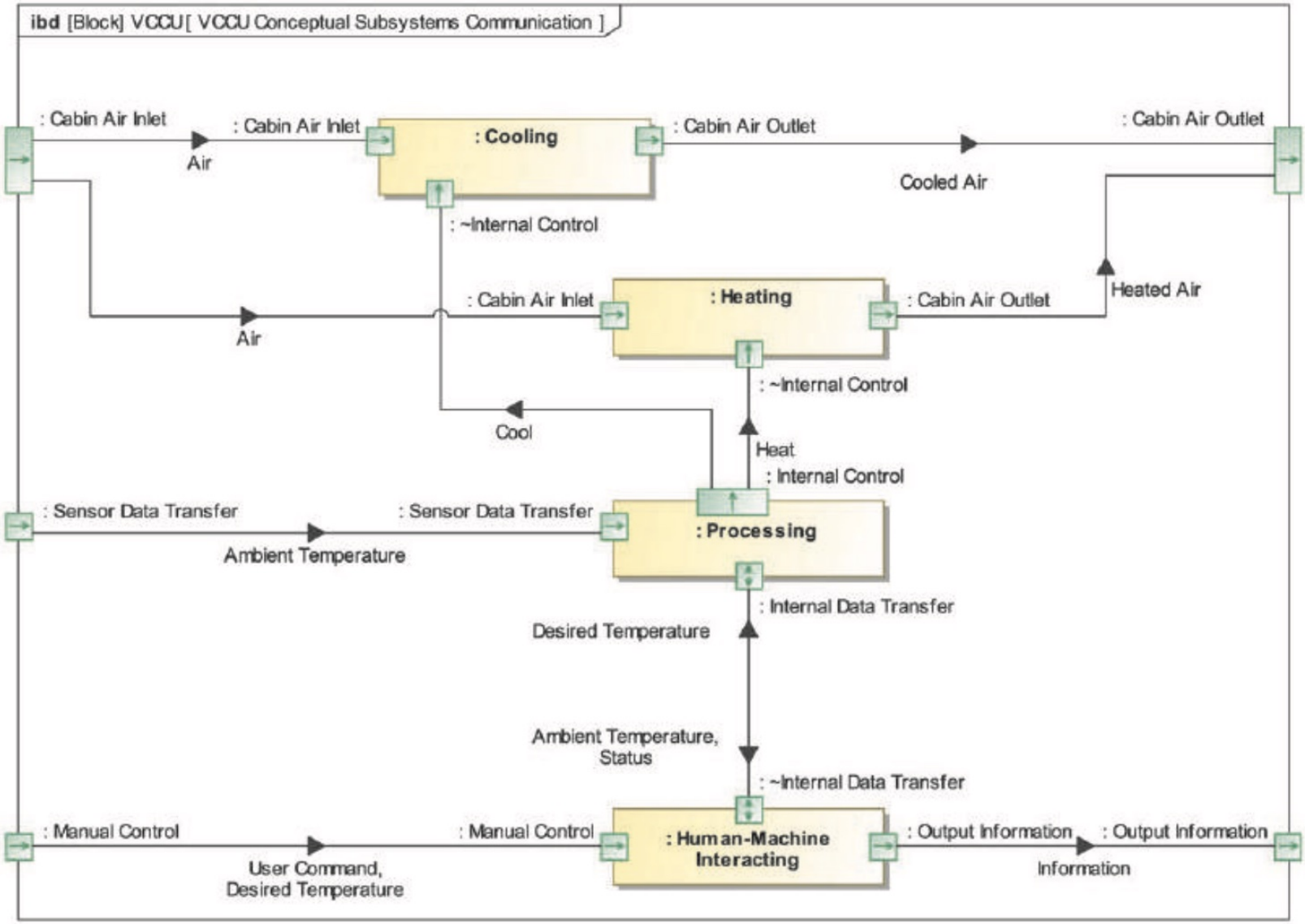
Example

- White-Box Perspective: Conceptual Subsystems



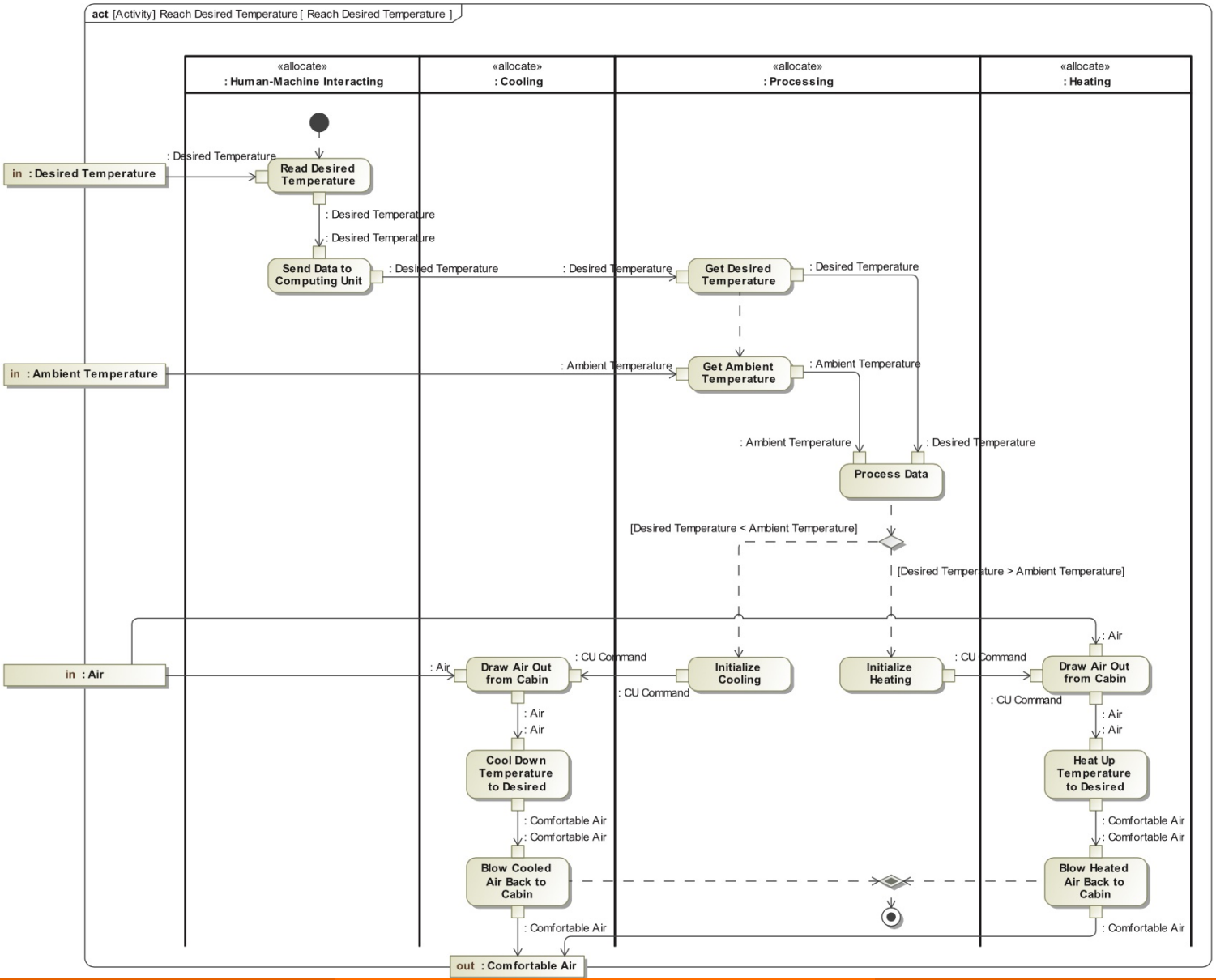
Example

- White-Box Perspective: Interactions among Subsystems



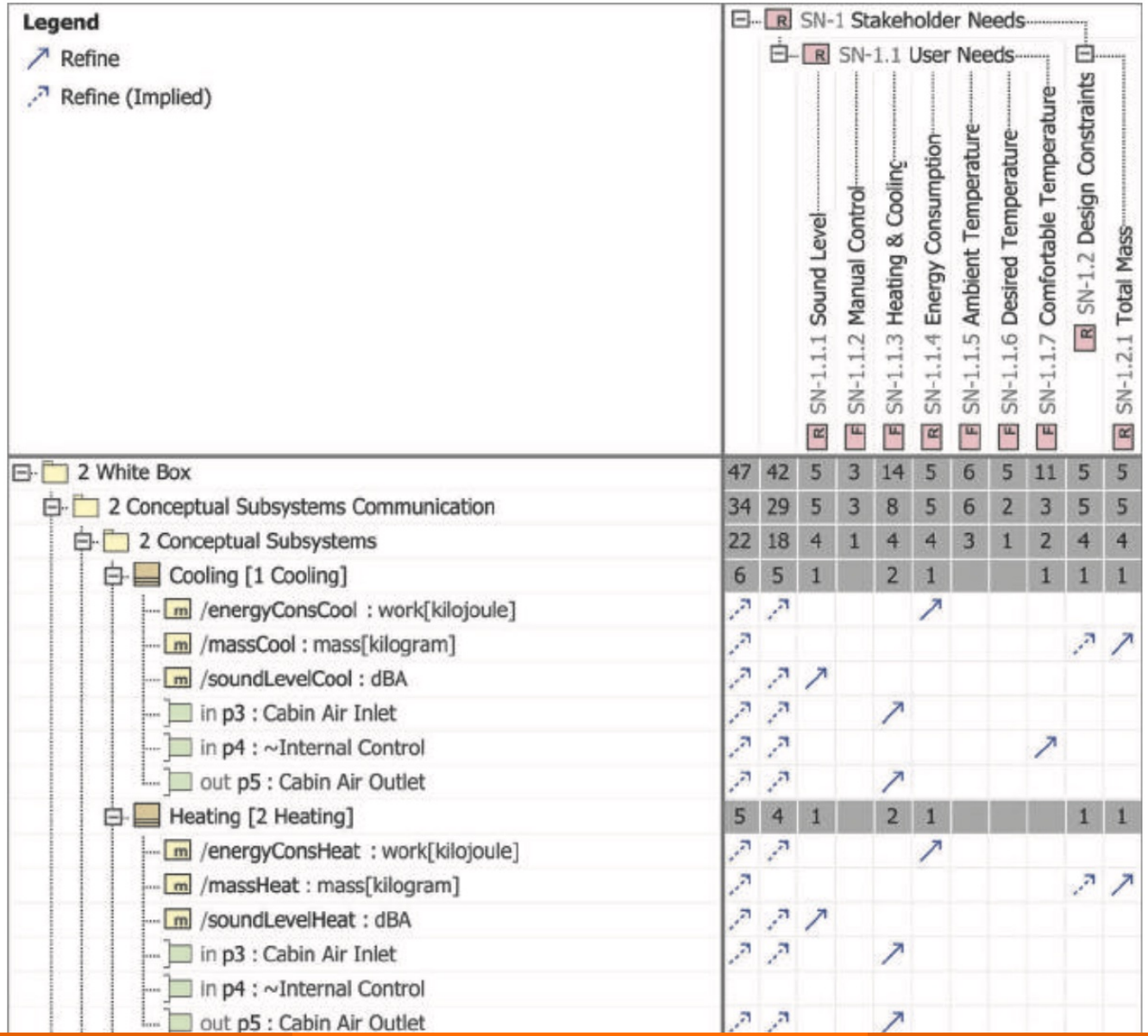
Example

- White-Box Perspective: Allocating Functions to Subsystems



Example

- White-Box Perspective: Creating a matrix for capturing refine relationships



Domain Specific Modeling Environments

Both UML and SysML are stantards, but both may be complex to understand (i.e., huge metamodels), let alone master.

- A practical approach is the develop Domain Specific Modeling Environments.
- An example is Mbeddr (https://mbeddr.com/), which is developed using MPS (https://www.jetbrains.com/mps/).

In RAMDE classes we follow this approach as a way to better understand modeling and all its implications.